David Mayowa Onaiyekan

AI Engineer

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Profile

AI Engineer (M.Sc. Artificial Intelligence) focused on making LLM systems cheaper without making the answers worse. Fine-tuned Qwen3-4B (QLoRA) to match Claude Sonnet 4.6 triage accuracy at about 17x lower cost per ticket, and built agent pipelines with human-in-the-loop approval, including one that handles messages in fewer Claude API calls. Built evaluation suites for accuracy and latency.

Skills

- **Languages:** Python (Advanced), JavaScript, TypeScript
- **Gen AI & LLM Engineering:** GPT, Claude, Gemini, LlamaIndex, Hugging Face, LangChain, LangGraph, CrewAI, RAG, MCP (FastMCP), Tool Calling, Structured Outputs (Pydantic), Prompt Engineering
- **Databases & Vector Databases:** PostgreSQL, MySQL, MongoDB, Qdrant, Pinecone, ChromaDB
- **Web & Full-Stack:** React, Next.js, Node.js, REST APIs, Prisma ORM, Tailwind CSS
- **ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **Computer Vision:** OpenCV, Image Processing, CNNs, Vision Transformers, OCR, GANs
- **MLOps, Cloud & Deployment:** AWS, Docker, FastAPI, CI/CD, GitHub Actions
- **Automation & Workflows:** n8n, Make, AI Workflow Orchestration

Education

Master of Science in Artificial Intelligence **01/2026**
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany

Bachelor of Technology in Computer Science **11/2018**
Federal University of Technology, Minna, Nigeria

AI Engineering Projects

Fine-tuning a Small LLM for Ticket Triage (Qwen3-4B, QLoRA) **09/2026 – 10/2026**

- Fine-tuned Qwen3-4B with QLoRA for German/English support-ticket triage, with the goal of matching Claude Sonnet 4.6 accuracy at a fraction of the cost.
- Benchmarked against my production Claude Sonnet 4.6 triage call on 25 real tickets: core accuracy improved by 20%, cost fell about 17x ($0.29 vs $5.11 per 1K tickets), with p50 latency 2.95s vs 2.08s.
- Tracked training runs in Weights & Biases and served the merged model with vLLM, containerized with Docker and FastAPI.

Tech: Python, PyTorch, Hugging Face, QLoRA, vLLM, FastAPI, Docker, Weights & Biases

Smart Dispatcher — Agentic Ticket Triage & Dispatch **04/2026 – 09/2026**

- Built a multi-step LangGraph agent that triages support tickets and drafts dispatch plans, with a human-in-the-loop checkpoint where a reviewer approves or edits every plan before it is executed.
- Implemented deterministic guardrails (regex-based part extraction and validation) that check the LLM's output against hard rules before any action is proposed.
- Exposed inventory, manuals and technician lookup as FastMCP tools, so the agent selects and chains tools per request through a composable, reusable tool layer.
- Added a self-correction loop that detects failed repair plans and retrieves firmware workarounds via RAG, recovering from 100% of the failures in testing.
- Built a 17-case evaluation suite measuring retrieval quality and task success rate, covering metadata-filtered vector search and the guardrails.

- Shipped a React/FastAPI dashboard that streams each agent step over SSE, with every run traced in LangSmith for observability and auditability.

Tech: Python, LangGraph, FastMCP, ChromaDB, FastAPI, React, LangSmith, Docker, Railway

RAG-Powered Document Chat System **02/2026 – 06/2026**

- Developed a publicly deployed multimodal RAG application enabling users to upload PDFs and receive streaming, grounded answers with page-level citations.
- Engineered a hybrid retrieval system combining dense BGE embeddings and BM25 with Reciprocal Rank Fusion and LLM-based reranking to optimize context retrieval across text and image chunks.
- Streamed live evaluation metrics—Hit Rate, MRR, faithfulness, answer relevancy, and correctness directly into the UI.

Tech: Python, FastAPI, LlamaIndex, SSE streaming, RAG, Qdrant, Docker, Railway, GitHub CI/CD

AI Ticket Bridge Agent — Teams to Jira Automation **02/2026 – 04/2026**

- Developed an AI bridge agent that classifies Microsoft Teams conversations, enriches them with CRM context, and reduces manual ticket creation to a one-click approval workflow.
- Optimized a multi-stage agent pipeline to process 70+ support messages using fewer than four Claude API calls while streaming progress to reviewers in real time.
- Built a review dashboard enabling human validation of AI recommendations before synchronizing approved actions with Jira and Microsoft Teams.

Tech: Python, Claude, Jira REST API, Docker, Git

Research Experience

Machine Learning Researcher **06/2025 – 01/2026**
Pattern Recognition Lab, FAU *Erlangen, Germany*

- Fine-tuned and evaluated alternative backbone architectures for the AFFGAN handwriting style encoder to better capture writer-specific features, benchmarking against the original baseline.
- Extended the system beyond word-level synthesis to line-level handwriting generation, integrating and adapting an existing line-generation architecture to support full-sentence output.
- Deployed the handwriting generation pipeline as a web application, enabling users to select a writing style, input custom text, and instantly generate a corresponding handwriting image.
- Integrated TrOCR into a teacher–student distillation framework to evaluate and improve readability of generated handwriting, improving OCR accuracy by 20% over the baseline.

Tech: PyTorch, Python, GANs, CNN, ViT, TrOCR, Streamlit, CUDA

Computer Vision Researcher **12/2023 – 01/2025**
Institute for Factory Automation and Production Systems (FAPS) *Erlangen, Germany*

- Collaborated within a research team to execute systematic data cleaning on thousands of industrial coil-winding images — manually correcting label noise on a representative subset and leveraging Cleanlab Studio's autofix to propagate corrections across the remaining dataset, establishing a high-fidelity baseline for downstream model training.
- Trained and evaluated fully-supervised baseline classifiers (DINOv2-L and EfficientNetV2s) to establish reference macro-F1 performance, benchmarking against SSL (FixMatch/MixMatch) gains across four label-fraction splits (10%, 25%, 50%, 100%).
- Ran systematic hyperparameter optimization with Optuna across backbone/algorithm combinations, tuning learning rate, batch size, and augmentation parameters, and tracked all trials via a persistent SQLite results database for reproducibility.
- Designed and implemented a config-driven multi-label SSL training pipeline supporting both FixMatch and MixMatch algorithms, achieving a 90% macro-F1 average score with FixMatch.

Tech: PyTorch, DINOv2-L, EfficientNetV2-L, FixMatch, Optuna, WandB, Docker

Languages

English (Native) — German (Intermediate)